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Interest rate risk of Chinese commercial banks based on the GARCH-EVT model

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Interest rate market risk faced by China's commercial banks is increasing after the announcement that the interest rate marketisation is completed. This paper examines the Value-at-Risk, and statistical properties in the daily price return of Shanghai banks' overnight offered rate. The study applies two-stage approaches, combining GARCH-type models with extreme value theory. Firstly, the Markov regime switching model is used to test the regime states of the series. Secondly, the performance of different VaR models are examined. Results show that the extreme value approach estimates better at the 99% confidence level. The EGARCH-GED model is the most suitable of the employed GARCH-type models. The back-testing results support the idea that the approach used in this study is appropriate for improving commercial banks' daily risk management. This paper applies the GARCH-EVT method for interest rate measurement after China's interest rate marketisation and added regime analysis of interest rate. Suggested policy implications will help formulate policies that guide the activities of commercial banks in China.

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Introduction

Volatility in the interest rate market profoundly affects asset pricing and risk management. Therefore, an increasing number of works concern the prediction of interest rate market volatility. The Chinese market has received a lot of attention from the world due to its expansion and the rapid development of the country's environment, social and government (ESG) assets and ESG investments (Liu et al. 2023), the stock market (Cai et al. 2023) and capital market (Sun et al. 2023). Affected by the pandemic, the world economy is facing a new recession (Mazur et al. 2021; Štifanić et al. 2020). China experienced a nationwide and long-term lockdown in 2020 to prevent the spread of the pandemic (Li et al. 2021). The economic growth rate in that year was significantly reduced, and the country's asset price exhibited more volatility (Zhang et al. 2023). The increase in economic risks has brought drastic interest rate fluctuations to the Chinese market, where interest rates align with the world. Chinese commercial banks, still under the bank-oriented financial system, are facing interest rate risks (Wang and Zong 2020).

Research regarding commercial banks' interest rate fluctuation is limited even though the industry keeps expanding. The available studies have focused on the liberalization of interest rates that was implemented in China. However, this implementation has raised many concerns because interest rates are determined by the market. If the market decides interest rates for commercial banks, they face a higher risk. Therefore, better research and prediction of interest rate fluctuations are of great significance for commercial banks to manage and control interest rate risks, operate steadily, and reduce systematic financial risk. This study intends to fill this gap by exploring and providing a better interest rate fluctuation model that will help minimize commercial banks' risk. Filling this gap serves as the novelty or originality of this study. The liberalization of interest rates that has been implemented in China has given so much power to the market which is beyond the control of these banks and investors. It is worth noting that, China's interest rate liberalization reform is implemented to adapt to the global interest rate liberalization trend, better integrate into the world economy, and promote national financial liberalization (Huang and Ji 2017). While increasing the vitality of the financial market, it will also lead to greater fluctuations in interest rates (Yang et al. 2020). If this continues in the foreseeable future, the future of commercial banks will be unpredictable and put depositors, investors, and the nation at a higher risk. Due to these limitations in the market, Commercial banks that have been under rate control for a long time do not make sufficient preparations to avoid the increased interest rate risk. In that regard, there is a need for the development and adoption of a more reliable interest rate fluctuation model that can help minimize these risks. Therefore, this paper is dedicated to studying the interest rate risk faced by commercial banks after China's interest rate liberalization, looking for a reasonable interest rate fluctuation prediction model, making more accurate predictions of future interest rate fluctuations, and providing corresponding policy recommendations for risk management. The research seeks to answer the following research questions (1) What are the risks commercial banks face in light of interest rate fluctuations? (2) How can these risks be minimized using the interest rate fluctuation model? (3) What policies are needed to guide commercial banks in the face of interest rate risks?

By providing answers to these relevant questions, the study will fill various gaps left in the literature, make meaningful research contributions, and provide policy implications to policymakers for effective policy formulation. First, the paper will provide a strong foundation for future studies regarding interest rate fluctuations and risks. This will guide these studies in identifying the most appropriate interest rate fluctuation model in their research.

Also, the paper will expose major stakeholders to the various risks they face in the market. It will bring to light the vulnerability of the industry in the face of market-determined interest rates and provide them with a gateway to embrace solutions. Finally, the study will provide strategies and implications that are aimed at providing a better interest rate fluctuation model to assist policymakers in organizations and the country at large. When this model is identified and utilized, industries can minimize their risks and maximize returns simultaneously. In that regard, the commercial banks industry moves a step further from crashing. Through these contributions, Chinese commercial banks can better face the fluctuating interest rate market and achieve sustainable and steady operation and growth.

The remainder of the paper is structured as follows: Section two is the literature review. Section three describes the data and method. Section four provides the empirical results. Section five discusses the results and the sixth section concludes and provides recommendations.

Literature review

Value-at-risk (VaR) interest rates. As a type of market risk, interest rate risk is measured suitably using the concept of value-at-risk (VaR) (Xia et al. 2023). The role of VaR is to provide financial institutions with an estimate of the maximum possible loss at a given confidence level. In practical applications, choosing an appropriate method to obtain a reasonable estimate of VaR is the biggest problem. Pérignon and Smith (2010) observed that 73% of American international banks used historical simulations for their VaR estimates. Only a few financial institutions used more complex parametric models for their calculations. The historical simulation method's important premise is that history will repeat itself. The data used to measure VaR is historical data. When the financial market experiences extreme volatility, this method cannot precisely measure the risks faced by banks in the future.

To obtain an accurate measure of VaR, it is very important to predict future market fluctuations (Liu et al. 2022). Experience has shown a volatility clustering effect in financial markets: price fluctuations tend to be persistent, and large price fluctuations in the current period are often accompanied by large price fluctuations in the next period, as well as low volatility. The application of the GARCH model (Bollerslev 1986; Amirshahi and Lahmiri 2023) is very helpful in capturing the effect of volatility clustering. The model assumed at the beginning is a normal distribution of returns. This distribution assumption may not effectively capture the number and impact of extreme changes in asset prices, which often results in risk underestimation.

Expansion of GARCH model. Subsequently, scholars introduced more flexible distributions. The model was set to have the characteristics of skewness and thick tail, more consistent with the financial return sequence, expanding and innovating the original GARCH model. Such as Normal, Student's t, biased t (Hansen 1994), generalized error distribution GED (Nelson 1991), inverse Gaussian distributions (NIG, Barndorff-Nielsen 1997), α -stable and slowly stable distributions (Broda et al. 2013). The GARCH family models are constantly enriched, and their distributions are constantly innovating. In addition to the constantly proposed new yield distributions of the GARCH model, GARCH models suitable for different situations have been put forward. Exponential GARCH, EGARCH for short (Nelson, 1991), Threshold GARCH, also known as TGARCH (Zakoian 1994), and its similar GJR-GARCH (Glosten et al. 1993), by adding variables representing the different effects of positive and negative shocks,

effectively characterizing the leverage effect of returns. Moreover, APARCH (Ding et al. 1993), FIGARCH (Baillie et al. 1996), and MNGARCH (Haas et al. 2013) are suitable for analyzing volatility with long-term dependence. The HN-GARCH model (Heston and Nandi, 2000; Wang and Zhang, 2022; Escobar-Anel et al. 2022) is suitable for calculating options and portfolio volatility in derivatives markets. Multivariate GARCH models—BEKK (Engle and Kroner 1995) and DCC (Engle 2002, Lee and Kim 2021, Ke et al. 2021) are useful for dealing with fluctuations in the covariance matrix of asset returns.

Scholars also use the GARCH family models to estimate different financial market risks. Chkili et al. (2014), Zhang and Zhang (2016), and carried out GARCH-VaR estimates and forecasts for precious metal markets. Baur et al. (2018), Colon et al. (2021), Guo (2021), and others made volatility predictions on the Bitcoin market with high volatility and active trading. Herwartz (2017) and used various GARCH models to forecast global stock market returns. However, using GARCH family models requires estimation based on the assumption of the return distribution, which has certain limitations. Some scholars use various GARCH family models to fit the data, reduce the fitting error caused by the model's active selection preference and have achieved certain results (Wang et al. 2023). However, this trial-and-error approach does not eliminate the model selection problem.

Extreme value theory (EVT). Applying extreme value theory (EVT) can resolve the difficult problem of tail model selection. The charm of extreme value theory is that when focusing on the tail distribution, it is possible to obtain the parametric form of the tail event without knowing the original distribution (Tomlinson et al. 2023). Extreme value theory provides a theoretical and practical basis for statistical models describing extreme events and is rapidly applied to measure financial value at risk. Previously, there has been substantial literature on extreme value theory and the tail distribution of assets (McNeil 1999, Longin 2000, Neftci 2000, Gilli and Kellezi 2006). Accurate estimation of the tail distribution can better measure risk exposure in extreme cases, to better manage assets. McNeil and Frey (2000) combined GARCH and EVT models to estimate the value at risk. The results show that the model captures the characteristics of the financial event sequence, adapts to the current market trend quickly, and can obtain a more accurate VaR estimate. Just (2014) confirmed that the GARCH-EVT method is also valid for estimating agricultural product markets.

Gaps and contributions. According to the literature reviewed, various studies have adopted the various GARCH models, EVT models, or both in their estimations. They have been applied and explored in various areas except the commercial bank industry. The study identifies this gap in the literature and uses the GARCH-EVT model to study the interest rate fluctuation in the commercial bank industry in China. This paper mainly studies the interest rate market risk faced by China's commercial banks after announcing that the interest rate marketisation is completed (Zhang et al. 2020). On October 24, 2015, the People's Bank of China decided to no longer set a floating ceiling on deposit interest rates for commercial banks and rural cooperative financial institutions, interest rates are dominated by the market. Overnight SHIBOR has the shortest term, the strongest liquidity, and the most sensitive response to the market. It is also the most used interest rate in all SHIBOR interest rate quotations in inter-bank market transactions. To estimate the interest rate risk of China's commercial banks, we use the extreme value-based GARCH-EVT method to estimate the VaR that China's

Table 1 Descriptive statistics and LB test results of O/N SHIBOR logarithmic first-order difference series.

Returns

Descriptive statistics	
Num.	1496
Mean	−0.0001
Std. dev.	0.1071
Maximum	1.050
Minimum	−0.488
Skewness	1.621
Kurtosis	14.908
LB-Q (5)	101.54 (<2.2e-16)
LB-Q (10)	112.29 (<2.2e-16)
LB-Qs (5)	118.14 (<2.2e-16)
LB-Qs (10)	141.04 (<2.2e-16)

Null hypothesis: The Series is not purely random.

Rejection of the null hypothesis that the test series is purely random at the 1% significant level.

commercial banks face after the interest rate marketisation. By filling this gap and making contributions, the VaR of these banks can be minimized to maximize banks' profitability and protect depositors' funds. Based on this, the study formulates this hypothesis.

Hypothesis: GARCH-EVT better predicts VaR that Commercial banks face after interest rate marketization

Data and method

Data sources. The data source used in this article is the Shanghai Bank Overnight Offered Rate (O/N SHIBOR), with 1497 transaction data from October 24, 2015, to October 22, 2021. As the basis for the pricing of deposit interest rates, SHIBOR is widely used in market-based product pricing and plays an irreplaceable benchmark role in China's financial market. Overnight SHIBOR is the most used interest rate in all SHIBOR interest rate quotations in inter-bank market transactions. Data can be obtained at the following website (<http://www.shibor.org/>).

Financial time series usually have the characteristics of sharp peaks and thick tails, so the first-order logarithmic difference is taken for the overnight SHIBOR return series to eliminate the non-stationarity of the original time series (Lu et al. 2023).

$$R_t = \ln \frac{r_t}{r_{t-1}} \quad (1)$$

A preliminary analysis was carried out, and it found that the volatility of SHIBOR increased after 2018Q2. Table 1 is the descriptive statistics and Ljung-Box (LB) test results of the log first difference of O/N SHIBOR. The LB test is a commonly used method for testing sequence autocorrelation in time series analysis. The statistics in Table 1 show that the logarithmic first-order difference fraction of O/N SHIBOR has the characteristics of a traditional financial time series with sharp peaks and thick tails. The null hypothesis is that the series is not purely random and cannot satisfy the establishment of the GARCH model. However, the correlation LB test shows that for the 5th and 10th order lag terms of LB statistics and LB square statistics, the test *p* values are all significant, satisfying the establishment of the GARCH model.

Research method. As a type of market risk, interest rate risk has been measured and predicted from the interest rate sensitivity gap management model, the duration model, and the expected shortfall model to the current main VaR method.

Markov regime switching model. The Markov regime switching (MS) model can analyze the time series with structural breaks or different regime states (Lin et al. 2021). To study the impact of interest rate liberalization, we use the Markov regime switching model proposed by Hamilton (1989) to explore the change in interest rates. A two-regime structure is demonstrated by high and low volatility states. The general form of the two regimes is as follows:

$$\varepsilon_t \sim \phi(\mu_k, \sigma_k) = p_{11}\xi_{k=1,t} + p_{22}\xi_{k=2,t} \quad (2)$$

Where p_{ij} is the transition probabilities, $\xi_{k,t}$ is the smoothed regime probabilities. The transition probabilities governed the regime switching process:

$$p_{ij} = P(S_t = j | S_{t-1} = i) \quad (3)$$

The probability transition matrix can be represented as follows:

$$\begin{bmatrix} p_{11} = 1 - p & p_{12} = p \\ p_{21} = q & p_{22} = 1 - q \end{bmatrix}$$

Where p gives the probability that regime 1 will be followed by regime 2. q gives the probability that regime 2 will be followed by regime 1. $1-p$ and $1-q$ gives the probability that there will be no change in the state of regime 1 or regime 2 in the following period. θ is the estimated parameter by maximizing the logarithmic likelihood function:

$$\max \ln L = \sum_{i=1}^T \ln f(p; \theta) \quad (4)$$

Definition and calculation formula of VaR. Xia et al. (2023) define Value at Risk (VaR) as given a confidence level, the maximum possible loss at this determined level.

$$P(\Delta V \leq VaR) = \alpha \quad (5)$$

ΔV is the amount of change in the value of the target asset

Using the parametric method, VaR values under different distributions can be calculated.

$$VaR_{\alpha,t} = \hat{\mu}_t + z_{\alpha} \hat{\sigma}_t \quad (6)$$

Among them, z_{α} is the quantile at the α significance level, and the left-end loss is measured by the formula.

Using ARMA-GARCH models to estimate mean and volatility. The Autoregressive Moving Average (ARMA) model is widely used to conduct time series analysis and prediction. If the dataset rejects the stationary hypothesis, this shows that the series is stationary and ARMA should be used to conduct the prediction (Yue et al. 2020). The GARCH model simulates the variance of random error terms, which is suitable for time series volatility analysis and can effectively eliminate excessive peaks in data fluctuations (Wang and Yan 2022). GARCH (1,1) is a model most suitable for modeling most financial time series.

The mean equation of the GARCH family model is:

$$y_t = \mu + \varepsilon_t \quad (7)$$

Among them, μ is the mean, ε_t is the residual term.

Standard GARCH(1,1) model: The standard GARCH(1,1) conditional variance term proposed by Bollerslev (1986) is

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8)$$

In the formula, if $\omega > 0$, $\alpha > 0$, $\beta > 0$ and $\alpha + \beta < 1$, then the conditional variance term is considered to be stationary.

EGARCH(1,1) model: Nelson (1991) proposes the model to capture the impact of asymmetric volatility caused by leverage effects. The leverage effect means that volatility behaves

differently in the face of good and bad news: after bad news comes out, volatility increases more than good news. The conditional variance term is

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \alpha \frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} \quad (9)$$

In the formula, α measures the leverage effect, and taking the logarithm of the variance ensures that the parameters in the formula have no value restrictions.

TGARCH(1,1) model: Zakoian (1994) proposed another model measuring the leverage effect. The method of piecewise function was used in the variance term, and the conditional variance term was

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma \varepsilon_{t-1} I_{t-1} \quad (10)$$

The formula satisfies $\omega > 0$, $\alpha + \gamma > 0$, $\beta > 0$, where I_{t-1} is a piecewise function, and the value is 1 when $\varepsilon_{t-1} < 0$, and 0 otherwise. If $\alpha + \beta + \frac{1}{2}\gamma < 1$, then the process is covariance stationary. If the coefficient of γ is significantly different from 0, we believe that the shock of good and bad news has an asymmetric effect on the conditional variance.

Three kinds of complex distributions used in parameter fitting: The probability density function of the GED distribution was proposed by Nelson, and the population obeys a probability density function with mean μ :

$$f_{GED}(x) = \frac{v e^{-|\frac{x-\mu}{\sigma}|^v}}{2\sigma\Gamma(1/v)} \quad (11)$$

Among them, $\Gamma(\cdot)$ denotes the gamma function, $-\infty < \mu < +\infty$ is the position parameter, $\sigma > 0$ is the shape parameter, $v > 0$ is the parameter for judging the thickness of the tail, and the GED distribution is normal distribution when $v = 2$, when $v < 2$, its density has thicker tails and sharper peaks than the normal distribution. The smaller the v , the more obvious the phenomenon of thick tails is, and the tails are thinner than the normal distribution when $v > 2$.

The probability density function of the SGED distribution:

By transforming the new GED distribution form proposed by Nadarajah (2005), the probability density function of the SGED distribution can be obtained:

$$f_{SGED}(x) = \frac{1}{\lambda + \frac{1}{\lambda}} \cdot \frac{v}{\sigma\Gamma(1/v)} \cdot \left[e^{-|\frac{x-\mu}{\sigma}|^v} I_{(0,\infty)}(x-\mu) + e^{-|\frac{x-\mu}{\sigma}|^v} I_{(-\infty,0)}(x-\mu) \right] \quad (12)$$

Among them, $I_{(\cdot)}(x)$ is the pointing function, λ is the skewness parameter; the function is skewed to the right when $\lambda > 1$, and the function is skewed to the left when $0 < \lambda < 1$

The probability density function of the NIG distribution:

The NIG distribution, first proposed by Barndorff-Nielsen, was originally used to model speculative returns. Its density function is:

$$f_{NIG}(x) = \frac{\alpha \delta}{\pi \sqrt{\delta^2 + (x^2 - \mu^2)}} \cdot e^{\delta \sqrt{\alpha^2 - \beta^2} + \beta(x-\mu)} \cdot K_1 \left(\alpha \sqrt{\delta^2 + (x^2 - \mu^2)} \right) \quad (13)$$

Among them, the parameter α determines the kurtosis of the distribution, β determines the degree of asymmetry of the distribution, and μ and δ determine the location and scale of the residue, respectively. A nice property of the NIG distribution is that the convolution is closed for fixed α and β . The daily returns follow the NIG distribution, and the daily returns for x days still follow the NIG distribution.

Calculate VaR using extreme value theory. The mathematical basis of extreme value theory (EVT) is the classical extreme value limit theory. The core idea of EVT is that the extreme tails of large-scale distributions can be approximated by the generalized Pareto distribution (GPD). For values above the threshold μ , we define their distribution as:

$$F_{\mu}(y) = P(X - \mu \leq x | X > \mu) = \frac{F(y + \mu) - F(\mu)}{1 - F(\mu)} \quad (14)$$

where $0 < y < x_0 - \mu$, $x_0 \leq \infty$.

The Pickands-Balkema-de Haan theorem is fundamental in POT, which states that for a large fundamental distribution F , there exists a function $\beta(\mu)$ such that:

$$\lim_{\mu \rightarrow x_0} \sup_{0 < y < x_0 - \mu} |F_{\mu}(y) - G_{\xi, \beta(\mu)}(y)| = 0 \quad (15)$$

$G_{\xi, \beta(\mu)}(y)$ is a generalized Pareto distribution with the following distribution function:

$$G_{\xi, \beta(\mu)}(y) = \begin{cases} 1 - \left(1 + \xi \frac{y}{\beta}\right)^{-\frac{1}{\xi}}, & \xi \neq 0, \\ 1 - \exp\left(-\frac{y}{\beta}\right), & \xi = 0, \end{cases} \quad (16)$$

When $\xi \geq 0$, $\beta > 0$, $y \geq 0$; when $\xi < 0$, $0 \leq y \leq -\frac{\beta}{\xi}$.

β is the scale parameter, and ξ is the shape parameter. Thick-tailed distributions have $\xi > 0$, while thin-tailed distributions have $\xi = 0$. A distribution with a finite right endpoint will be $\xi < 0$.

The cumulative distribution estimated with the actual in-sample is:

$$\hat{F}(\mu) = 1 - \frac{N_{\mu}}{n} \left(1 + \hat{\xi} \frac{(x - \mu)}{\hat{\beta}}\right)^{-\frac{1}{\hat{\xi}}} \quad (17)$$

For the left end VaR, it can be estimated by the following equation:

$$VaR_{\alpha} = \mu_{t+1} + \sigma_{t+1} \times \left(\mu + \frac{\hat{\beta}}{\hat{\xi}} \left(\left(\frac{n}{N_{\mu}} \alpha\right)^{-\hat{\xi}} - 1\right)\right) \quad (18)$$

Back-testing. The back-testing is to compare the VaR calculated by the model with the actual profit and loss to test the accuracy and reliability of the model and improve and optimize the model accordingly.

Likelihood ratio unconditional coverage test by Kupiec. The Kupiec (1995) test holds that, for a valid estimation model, the proportion of the estimated VaR that is not reached should be comparable to the level of tail probability of the data. This test is mainly used to judge whether the fitted model can correctly estimate the risk of tail VaR and the difference between the number of risk days estimated and the actual number of days. If the total number of days in the sample is n and the estimated value is not reached for N days, then the failure frequency is N/n . The null hypothesis of the Kupiec test is $H_0: N/n = \alpha$.

Statistic function:

$$LR = 2 \ln \left(\left(1 - \frac{N}{n}\right)^{n-N} \left(\frac{N}{n}\right)^N \right) - 2 \ln((1 - \alpha)^{n-N} \alpha^N) \quad (19)$$

Under the null hypothesis, the LR statistic asymptotically follows a chi-square distribution with 1 degree of freedom. If the p value of the unconditional coverage test is less than a predetermined significance level, we reject the null hypothesis.

Christoffersen's Independence Test. The Kupiec test only focuses on the number of non-conforming VaRs and does not consider

Table 2 Markov regime switch model estimation.

Transition probabilities		Regime	
p_{11}	p_{21}	θ_1	θ_2
0.9482	0.0518	0.0007(-0.1628)	7e-04*(1.75)
AIC	-4642.399		

when extreme VaR values occur. Christoffersen's (1998) test was used to test whether abnormal VaR values were clustered together. The basic idea is to compare the probability of a violation occurring the next day when a violation occurs on the same day and the probability of a violation occurring the next day when the violation does not occur on the same day.

LR Statistics:

$$LR_C = 2 \ln((1 - p_{01})^{n_{0,0}} p_{01}^{n_{0,1}} (1 - p_{11})^{n_{1,0}} p_{11}^{n_{1,1}}) - 2 \ln(p^N (1 - p)^{n-N}) \quad (20)$$

Among them, $p_{ij} = p(I_t = j | I_{t-1} = i)$, indicating whether VaR conforms to the function estimated by the model for two consecutive days. If there is no model violation, then $I_t = 0$, otherwise $I_t = 1$. $n_{i,j}$ represents the number of days that state j occurs one day after state i occurs in the back-test sample. $p = \frac{N}{n}$.

If the test p value is less than the significance level, the null hypothesis is rejected. The VaR forecast is valid when the default sequence meets the unconditional coverage and independence assumption.

We will perform the above two back-tests on the GARCH-EVT model estimates to judge the validity of the VaR estimates.

Empirical results

Firstly, we perform regime switch estimation on the SHIBOR series and get the as presented in Table 2.

From the estimated results of the model, after the marketisation of interest rates, SHIBOR has shown an obvious state transition. Taking 2018 as the dividing line, it was in a state of low volatility for a long time before and then turned to a state of high volatility. Moreover, the probability of the two-state Markov transition matrix shows that a low-volatility stage is more likely to shift to a low-volatility stage than a high-volatility stage; and when the interest rate is in a high-volatility state, it is more likely to shift to a high-volatility stage. The marketisation of interest rates has increased the volatility of SHIBOR. This can be seen in Fig. 1.

By analyzing the nature of the previous data and observing the autocorrelation coefficient, we found that the O/N SHIBOR return series in the past 6 years has unpredictability and strong autocorrelation. Autocorrelation is the correlation between the expected values of random error items. Partial autocorrelation is the relationship between observations in a time series and observations at previous time steps that remove the relationship between intervening observations (Li et al. 2017). To eliminate the effect of autocorrelation on log-differenced series, we used the AR(1) model after comprehensive observation of autocorrelation ACF and partial autocorrelation PACF plots. For such a series with fluctuating aggregation, we use the GARCH family model to fit it. Table 3 displays the O/N SHIBOR log-differenced return series using various distributions of AR(1)-GARCH(1,1). The parameter estimates are shown in Table 3:

The basis for selecting the best model is that the model has the smallest AIC value (Zhang et al. 2020). We can see that using the standard GARCH(1, 1) model for fitting the sged distribution performs best. In contrast, the standard distribution and the biased standard distribution do not describe the distribution of the series well. However, for the significance of the parameters,

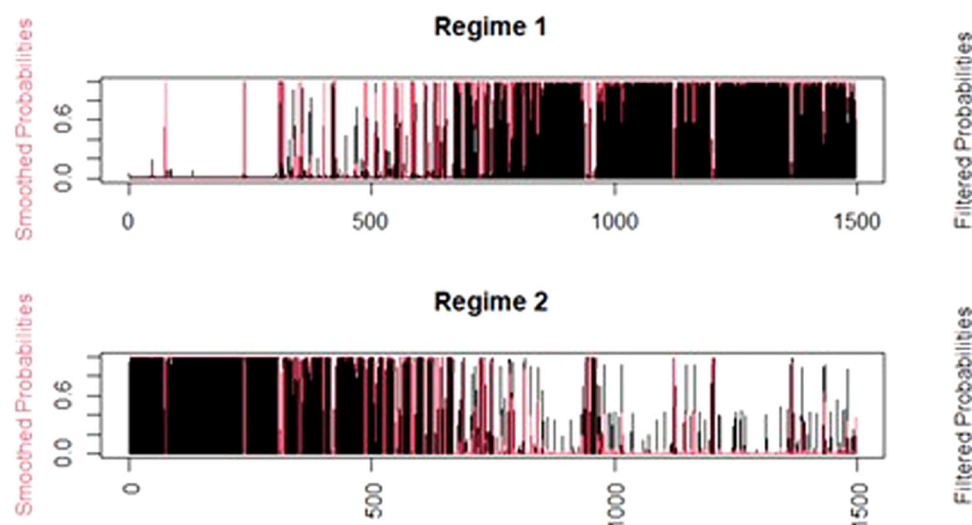


Fig. 1 The smoothed probabilities of two regimes of estimated model. Note: This figure is used to observe the market cycle and when the market “changes” during the selected sample time. The red line represents smoothed probabilities indicates the probability of remaining “unchanged” under the current regime. The closer it is to 1, the less likely the current regime is to change; the black line is the actual “filtered” probability.

Table 3 AR(1)-GARCH(1,1) parameter estimation.

Para meters	Garch-n	Garch-snorm	Garch-std	Garch-sstd	Garch-ged	Garch-sged	Garch-nig
μ	0.0020 (0.0057)	0.0028 (0.0045)	0.0006 (0.0372)	0.0008 (0.0186)	0.0002 (0.0000)	0.0004 (0.00000)	0.0010 (0.0007)
AR(1)	0.0148 (0.7493)	0.0442 (0.3416)	0.2681 (0.0000)	0.2758 (0.0000)	0.2710 (0.0000)	0.2741 (0.0000)	0.2954 (0.0000)
ω	0.0001 (0.1704)	0.0001 (0.1992)	0.0001 (0.8548)	0.0001 (0.8491)	0.0001 (0.6507)	0.0001 (0.5210)	0.0001 (0.7046)
α	0.2711 (0.0000)	0.2470 (0.0000)	0.3491 (0.0000)	0.3556 (0.0000)	0.2933 (0.0000)	0.2989 (0.0000)	0.3279 (0.0000)
β	0.7280 (0.0000)	0.7520 (0.0000)	0.6499 (0.0000)	0.6434 (0.0000)	0.7057 (0.0000)	0.7001 (0.0000)	0.6711 (0.0000)
Shape			3.7219 (0.0000)	3.7356 (0.0000)	0.7406 (0.0000)	0.7442 (0.0000)	0.3604 (0.0000)
Skew		1.1235 (0.0000)		1.0680 (0.0000)		1.0269 (0.0000)	0.1435 (0.0001)
AIC	-3.0231	-3.0413	-3.5224	-3.5267	-3.5571	-3.5604	-3.5394

even though the sged distribution is an optimal model, the coefficient ω is not significant, indicating that standard GARCH models and various distributions cannot fully describe the overall sequence.

Next, we use the EGARCH and TGARCH models with leverage to estimate the model's parameters. The results are shown in Tables 4 and 5:

Tables 4 and 5 show the fitting results of the EGARCH and TGARCH models, respectively. The α value of the EGARCH table is negative, and the γ of the TGARCH table is significantly different from 0, indicating that the overnight SHIBOR logarithmic return series has an asymmetric effect: the series falls more on bad news than it rises on good news. The lowest AIC values are EGARCH-GED and TGARCH-SGED. The tail thickness parameter is less than 2, indicating that the tail distribution of overnight SHIBOR returns is thinner than the Gaussian distribution. The model with the lowest AIC value in the three tables is EGARCH-GED, so we chose it as the model for VaR estimation. It can be seen from the parameter estimation that the value of parameter β exceeds 0.95 and reaches 0.965, indicating that the current fluctuation of more than 96% will affect the next period in the future. This data clearly illustrates the clustering effect of volatility and the slowness with which volatility shocks subside.

After selecting the best GARCH model, the autocorrelation and partial autocorrelation plot are shown next in Fig. 2.

In Fig. 2, the acf of the standardized residual is shown on the left and the acf of the square of the standardized residual is on the right. Standardized residual is the residual divided by the standard deviation. The model residual has eliminated the autocorrelation. The Ljung-Box test results are displayed in Table 6. According to that Table, the residual term of the model is not significant and can be considered white noise.

Then we use the obtained best estimation model to predict VaR. Table 7 shows the VaR estimates after 1 day based on the optimal model.

After obtaining the above calculation results using the GARCH family model, we use the extreme value method to calculate the VaR of O/N SHIBOR. We first make a hill diagram and a mean excess diagram and combine the two diagrams for observation.

Figure 3 shows the hill graph and mean excess point graph of the residuals of the EGARCH sequence. From this, we can see that the value of $\hat{\xi}$ is positive, which is in line with the thick-tailed property of the financial sequence. At the threshold of $\mu = 1.5853$, the original straight line appears to change significantly.

The estimation of the EVT method parameter and VaR can be seen in Table 8.

Table 4 AR(1)-EGARCH(1,1) parameter estimation.

Parameters	EGarch-n	EGarch-snorm	EGarch-std	EGarch-sstd	EGarch-ged	EGarch-sged	EGarch-nig
μ	0.0018 (0.0150)	0.0021 (0.0697)	0.0005 (0.0382)	0.0008 (0.0090)	0.0000 (0.0000)	0.0003 (0.0000)	0.0011 (0.0000)
AR(1)	0.0598 (0.0170)	0.0784 (0.0663)	0.2602 (0.0000)	0.2640 (0.0000)	0.2674 (0.0000)	0.2667 (0.0000)	0.2760 (0.0000)
ω	-0.2633 (0.0004)	-0.2182 (0.0003)	-0.0643 (0.0002)	-0.0032 (0.8253)	-0.2088 (0.0000)	-0.2017 (0.0000)	-0.1417 (0.0000)
α	-0.1884 (0.0000)	-0.1649 (0.0000)	-0.6739 (0.0000)	-1.8181 (0.0000)	-0.3000 (0.0000)	-0.2924 (0.0000)	-0.3184 (0.0000)
β	0.9429 (0.0000)	0.9504 (0.0000)	0.9797 (0.0000)	0.9863 (0.0000)	0.9634 (0.0000)	0.9654 (0.0000)	0.9747 (0.0000)
γ	0.5653 (0.0000)	0.5634 (0.0000)	1.3682 (0.0000)	3.8235 (0.0000)	0.6608 (0.0000)	0.6500 (0.0000)	0.7085 (0.0000)
Shape			2.1000 (0.0000)	2.0100 (0.0000)	0.6552 (0.0000)	0.6644 (0.0000)	0.1753 (0.0033)
Skew		1.1311 (0.0000)		1.0506 (0.0000)		1.0182 (0.0000)	0.1603 (0.0005)
AIC	-3.1346	-3.1509	-3.5914	-3.6040	-3.6188	-3.6175	-3.5928

Table 5 AR(1)- TGARCH (1,1) parameter estimation.

Parameters	tGarch-n	tGarch-snorm	tGarch-std	tGarch-sstd	tGarch-ged	tGarch-sged	tGarch-nig
μ	0.0014 (0.0277)	0.0020 (0.0149)	0.0007 (0.0174)	0.0008 (0.0096)	0.0002 (0.0000)	0.0004 (0.0000)	0.0010 (0.0000)
AR(1)	0.0258 (0.5381)	0.0322 (0.4590)	0.2540 (0.0000)	0.2606 (0.0000)	0.2664 (0.0000)	0.2753 (0.0000)	0.2858 (0.0000)
ω	0.0001 (0.1555)	0.0001 (0.1732)	0.0001 (0.8204)	0.0001 (0.8259)	0.0001 (0.5111)	0.0001 (0.4381)	0.0001 (0.0000)
α	0.1040 (0.0010)	0.1137 (0.0007)	0.2548 (0.0000)	0.2715 (0.0000)	0.2115 (0.0001)	0.2279 (0.0000)	0.2706 (0.0000)
β	0.7599 (0.0000)	0.7660 (0.0000)	0.6548 (0.0000)	0.6492 (0.0000)	0.7079 (0.0000)	0.7037 (0.0000)	0.6736 (0.0000)
γ	0.2702 (0.0000)	0.2344 (0.0000)	0.1789 (0.0147)	0.1528 (0.0383)	0.1591 (0.0199)	0.1299 (0.0000)	0.1043 (0.0000)
Shape			3.7974 (0.0000)	3.8019 (0.0000)	0.7510 (0.0000)	0.7531 (0.0000)	0.3730 (0.0000)
Skew		1.0685 (0.0000)		1.0505 (0.0000)		1.0243 (0.0000)	0.1224 (0.0000)
AIC	-3.0775	-3.0816	-3.5296	-3.5312	-3.5625	-3.5635	-3.5405

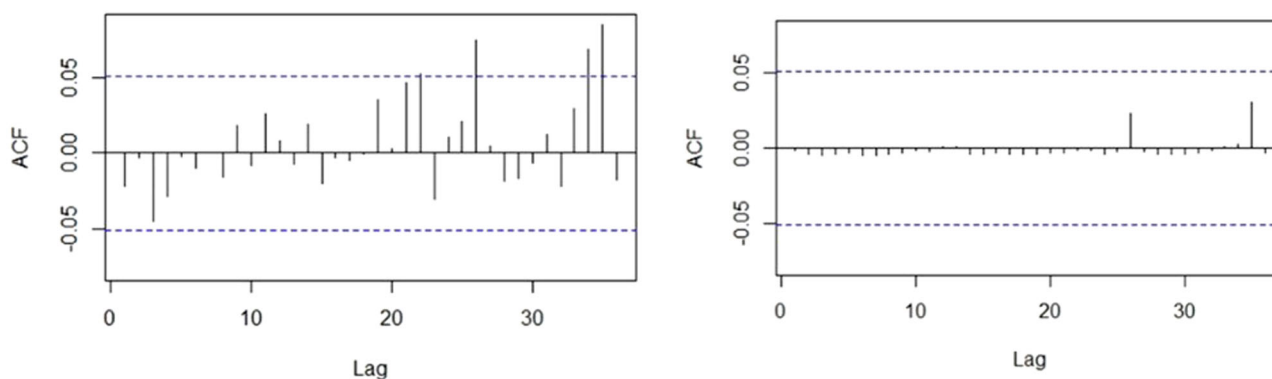


Fig. 2 Autocorrelation and partial autocorrelation plot of standardised residuals for O/N SHIBOR return. Note: Autocorrelation and partial autocorrelation plots. The vertical axis is the correlation coefficient, indicating the correlation of residual series; horizontal axis is the order of lag; the black vertical line represents the value of the autocorrelation coefficient of each lag. The blue area is the 95% confidence interval, which is the criterion for detecting whether the correlation coefficient is 0. When the black line exceeds these two boundaries, it can be determined that the correlation coefficient is significantly different from 0.

Table 6 LB Test on O/N SHIBOR standardized residuals.

LB-Q(5)	4.9918(0.4169)
LB-Q(10)	6.1274(0.8044)
LB-Qs(5)	0.0999(0.9998)
LB-Qs(10)	0.2124(1)

Table 7 Estimation of VaR after one day based on AR(1)-EGARCH(1,1)-GED model.

$VaR_{0.01}$	-0.5351
$VaR_{0.025}$	-0.3779
$VaR_{0.05}$	-0.2700
$VaR_{0.10}$	-0.1726

Comparing the VaR values calculated by the GARCH method and the EVT method under different confidence intervals, the following conclusions can be drawn. The GARCH-EVT method is used to estimate better the 1% extreme cases. At the 97.5, 95, and 90% confidence levels, the VaR calculated by the general GARCH family model is lower than the extreme value method. General GARCH models are good for estimates at low confidence levels.

In Fig. 4, the study made a moving window graph for the GARCH-EVT method and used the rolling model to fit continuously and get the results. It can be observed that choosing to

use different significance levels, the results of the VaR curves that are rolled out are quite different, with the lowest VaR breaking through -2% at the 1% significance level, while at the 10% significance level, this figure is less than -1%.

From top to bottom are 1, 5, and 10% of the moving window rolling VaR plot. The horizontal axis is the trading day, and the vertical axis is the loss value (%)

The model was back-tested using the Kupiec and Christoffersen backtesting methods. Table 9 shows the back-test table

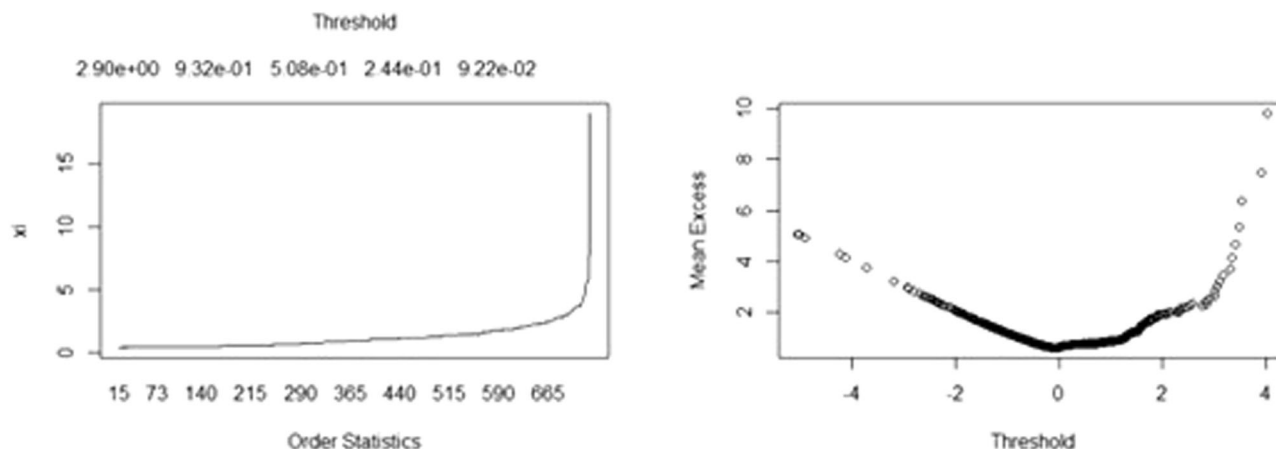


Fig. 3 Hill plot and mean excess plot. Note: Hill plot and mean excess plot used to determine the threshold. In the Hill plot, the upper horizontal axis is the threshold. The lower horizontal axis is the tail statistics arranged in order, and the vertical axis is the significance level. In the Mean excess plot, each open circle represents a tail statistic, the horizontal axis is the threshold, and the vertical axis is the mean excess level.

Table 8 EVT method parameter and VaR estimation.

μ	1.5853
N_{μ}	50
ξ	0.4791 (0.1921)
β	0.7289 (0.1667)
$VaR_{0.01}$	-0.6609
$VaR_{0.025}$	-0.3453
$VaR_{0.06}$	-0.2135
$VaR_{0.10}$	-0.1340

data. From this, it can be concluded that it has passed the significance test at the 99 and 95% confidence levels. Therefore, the null hypothesis cannot be rejected and proves that the setting of our model is more reasonable. The GARCH-EVT model captures the O/N SHIBOR fluctuations for a period, making it a great model for predictions.

The upper half of the back-testing table is Kupiec back-testing, and the lower half is Christoffersen back-testing.

Finally, a robustness test is performed. The method of changing the sample period was adopted to carry out the test. The test results are shown in Table 10.

The test results show that EGARCH-GED has the lowest AIC value and is the best fit, which is consistent with the previous estimated results.

Discussion

This work applied the GARCH model combined with the EVT model to estimate and predict the interest risk that commercial banks face. The results show that both the GARCH model and the GARCH-EVT model predict the interest risk more accurately in specific degrees of significance levels. The GARCH-EVT method estimates better in 1% of extreme cases. General GARCH models perform better at low confidence levels. This may be related to the fact that a single GARCH model ignores the accident of the series. The single GARCH is modeled in the entire period of the training set for a long time, while the volatility aggregation occurs quickly and suddenly. Therefore, this feature is difficult to reflect in the GARCH model. Our findings are not consistent with previous studies. The research of Ma et al. (2014) shows that the two-factor Vasicek model fits SHIBOR well, especially for SHIBOR in terms of three months or more. This

may be due to the difference caused by a different selection of the research period.

SHIBOR time series have characteristics of sharp peaks and thick tails. The series is not completely random based on the judgment of the LB test. The state transition and the increase of the volatility of SHIBOR, which turned from a low-volatility stage to a high-volatility stage show the impact of the marketisation of interest rates. Market decision mechanism means more dynamism, but also brings higher risks. Markov transition matrix shows that the volatility at present is affected by the past state and is likely to remain unchanged. In the future, high volatility of interest rates will become normal.

Conclusions and implications

Conclusion. This paper uses the GARCH family model to analyze the O/N SHIBOR quoted interest rate in the Chinese market after the interest rate liberalization is completed. The results show that O/N SHIBOR has the characteristics of a common financial time series. Sharp peaks and thick tails and behavioral characteristics of volatility clustering and leverage effect. We first use the GARCH model, EGARCH model, and TGARCH model, combined with 7 standard and biased distributions to fit the sequence; the results show that the AR(1)-EGARCH(1,1)-GED model is the most suitable for this SHIBOR interval. Next, we combined the GARCH model with the EVT method to estimate the VaR of value at risk and compared the results with the VaR calculated using the GARCH model alone. The research results show that at the 99% confidence level, the EVT method is estimated to be better than the general GARCH model, with a value at risk of 0.6609 or 0.6609%; at the 97.5, 95, and 90% confidence levels, the GARCH model calculation results are better. Finally, we performed moving windows and back-testing of the model, and the back-test results showed that the model results were significant. These results provide great insight into interest rate fluctuation risks faced by commercial banks and other sectors.

Policy implications. The study not only exposes the problems faced by commercial banks regarding interest rate fluctuations but also provides meaningful policy implications to aid policy-makers in minimizing the risks faced by commercial banks in China. First of all, the above empirical results show that the interest rate risk faced by commercial banks on the next day does not exceed 0.6609% of the bank's total assets under the 99% confidence level, and the single-day risk exposure faced by banks

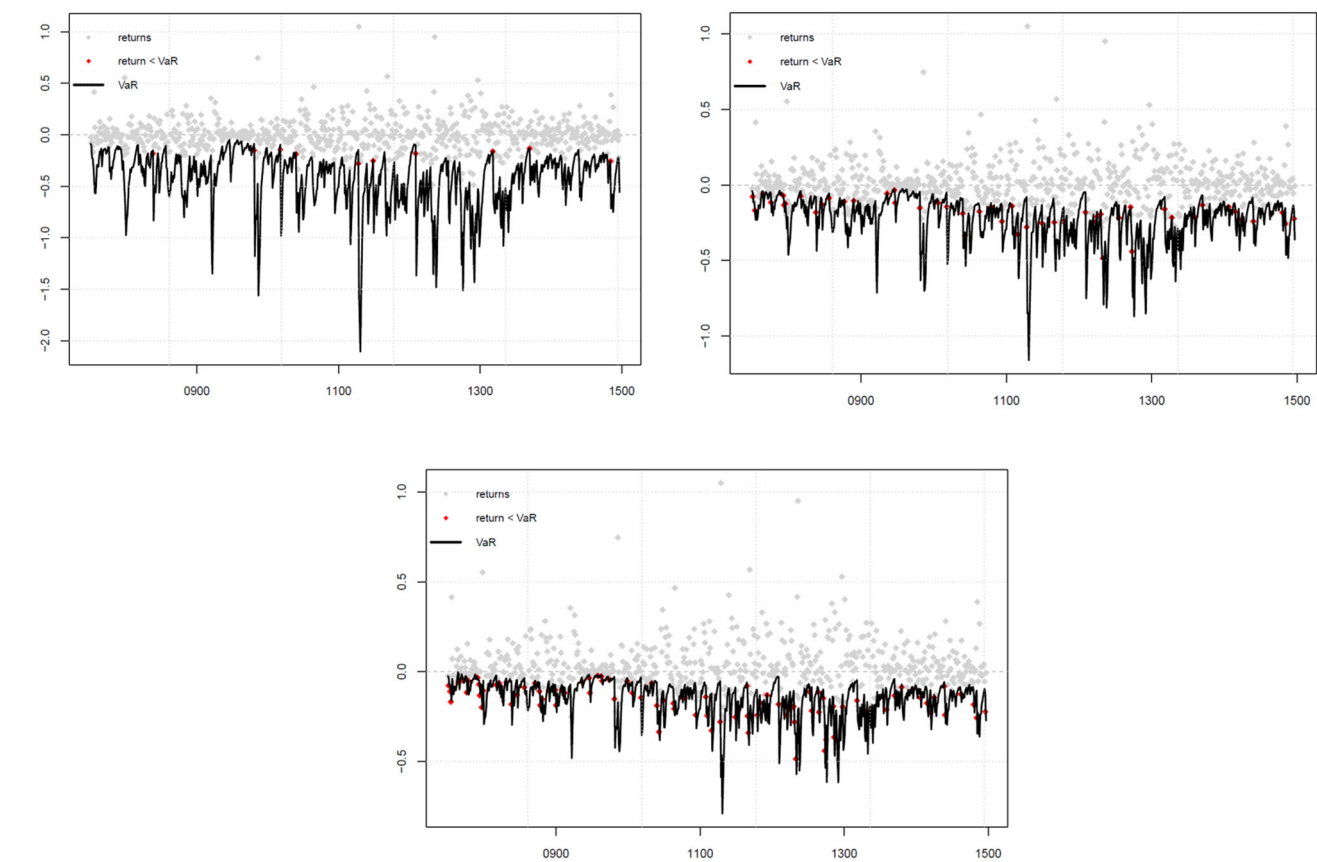


Fig. 4 1, 5, 10% moving window rolling VaR plot. Note: Moving window average VaR plot. The vertical axis is the rate of return (%), and the horizontal axis is the number of moving days. The gray points represent the actual return level, the continuous black line represents the VaR at the corresponding significance level, and the red dot represents the situation where the return is less than VaR at the corresponding significance level.

Table 9 Back-testing results.	
$LR_{0.01}$	0.776
EE/AE	7.5/10
Decision	Fail to reject H_0
$LR_{0.05}$	3.465
EE/AE	37.4/49
Decision	Fail to reject H_0
$LR_{0.1}$	9.294
EE/AE	74.8/101
Decision	Reject H_0
$LR_{C,0.01}$	1.047
Decision	Fail to reject H_0
$LR_{C,0.05}$	4.062
Decision	Fail to reject H_0
$LR_{C,0.1}$	9.306
Decision	Reject H_0

EE/AE means the ratio of expected exceed (EE) and the actual exceed (AE) of the tested data series.

is not large. However, risks will continue to accumulate, and the daily risk exposure will become a figure that cannot be ignored after some time. Banks still need to be cautious when managing interest rate risk in the future. The O/N SHIBOR interest rate is the benchmark interest rate in China’s capital market pricing products. Its fluctuations will be multiplied on multiple platforms, such as stock indices, commodities, and futures. Today, the business model of China’s commercial banks is gradually diversified. It is no longer limited to simple deposit and loan activities, and some businesses involve leveraged transactions.

Secondly, based on the impact of market-deterministic interest rates, commercial banks must first fully understand the importance of interest rate risk to commercial banks and strengthen their awareness of interest rate risk. Even though the study suggests an alternative way of dealing with interest rate fluctuations, it must be imperative for commercial policymakers to give a high priority to having an in-depth grasp of what the bank faces in the foreseeable future. Having a full grasp of the rate fluctuations is a great step towards controlling and minimizing it. Since the interest rate is interconnected with other activities regarding the commercial bank like depositors’ activities, authorities, and policymakers must create awareness among these stakeholders on the interest rate risks the bank faces. The banks cannot operate without risk; however, stakeholders’ interests must be a priority to help them minimize the risk and have fruitful banking.

Moreover, the study believes that risks can only be minimized, not eradicated by commercial banks because their success and failure depend on it. According to the findings of this study in minimizing the interest rate fluctuations for commercial banks in China, the establishment of product and pricing mechanisms become key to the sector. Commercial banks should establish a complete financial product pricing mechanism, set up a full-time financial product pricing department, and separate traditional financial product marketing from pricing. This will benefit all stakeholders to minimize the risk associated with interest rate fluctuation that is almost uncontrollable. When a complete financial product pricing mechanism is established, it can offset some of the unavoidable risks the banks face. This can be achieved effectively through the establishment of a full-time

Table 10 Robustness test results.

Para meters	Garch-n	Garch-snorm	Garch-std	Garch-sstd	Garch-ged	Garch-sged	Garch-nig
AIC	-2.2050	-2.2198	-2.5542	-2.5549	-2.6089	-2.6094	-2.5837
Parameters	EGarch-n	EGarch-snorm	EGarch-std	EGarch-sstd	EGarch-ged	EGarch-sged	EGarch-nig
AIC	-2.4065	-2.4072	-2.6645	-2.6718	-2.6958	-2.6954	-2.6727
Parameters	tGarch-n	tGarch-snorm	tGarch-std	tGarch-sstd	tGarch-ged	tGarch-sged	tGarch-nig
AIC	-2.3288	-2.3280	-2.5782	-2.5769	-2.6314	-2.6293	-2.6013

department to oversee these activities. This is a serious issue and should be given the necessary attention to change the face of interest rate fluctuation risks commercial banks face in the country. Another department that can be established is a full-time interest rate risk management department to meet the needs of commercial bank operations and management. In that case, full attention can be given to interest rates by the department to minimize risks.

Finally, commercial banks should incorporate interest rate risk into the scope of business training and regularly organize training for interest rate risk management personnel. Commercial banks can hire professionals and university professors to train employees to master knowledge of interest rate risk management and interest rate risk measurement methods. Besides, commercial banks should pay attention to and measure interest rate risk promptly to control risk exposure better and ensure the smooth operation of the bank. Based on the current data, it is predicted that commercial banks should maintain a liquidity reserve of more than 1% in a considerable window, to have sufficient funds to offset risk exposures. The prediction results of this paper can also provide a reference for policymakers.

Limitations of the study. However, this research has some limitations. With the rapid development of the economy and society, the changes in the volatility of time series in the future are difficult to predict. The model and the prediction cannot be precise anymore. More GARCH models may be used for estimation. Multi-regime analysis may be applied to get thorough results. Conditional value at risk (CVaR) measures the expected tail loss of financial assets beyond VaR, which is a good complement to measure extreme risk. In addition, we are also interested in the transmission of the impact of fluctuations in the benchmark SHIBOR interest rate on the impact of interest rate fluctuations on financial products invested by banks, which is the direction of our further research in the future.

Data availability

The datasets generated during and/or analyzed during the current study are available in the SHIBOR repository, and can be obtained at (<http://www.shibor.org/>).

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Author contributions

Conceptualization, DT and ZS; methodology, ZS; software, ZS; validation, DT and ZS; formal analysis, DT, ZS, XC, BZ and VB; investigation, ZS; resources, ZS; data curation, DT; writing—original draft preparation, XC and ZS; writing—review and editing, XC, DT, ZS, BZ and VB; visualization, VB and XC.; supervision, XC and DT; project administration, DT.

Competing interests

The authors declare no competing interests.

Additional information

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